COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE007	**SECTION:**	CPE11S4
DATE PERFORMED:	10/14/2026	**DATE SUBMITTED:**	10/16/2026
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ACTIVITY NO. 6.1 Functions

1. PROCEDURE OUTPUT

What procedures did you perform?

Perform each of these steps for the below problems:
Create a program in C++ that will add, subtract, divide, multiply for a 2 user-input integer values. Use functions in the program.

1. Write the functions:

```cpp
#include <iostream>
using namespace std;

// Returns the sum of two integers
int add(int a, int b) {
    return a + b;
}
```

```cpp
// Returns the difference of two integers
int subtract(int a, int b) {
    return a - b;
}

// Returns the product of two integers
int multiply(int a, int b) {
    return a * b;
}
```

```cpp
// Returns the quotient as a decimal value
double divide(int a, int b) {
    return (double) a / b;
}
```

2. Write the main() function:

```cpp
int main() {
    int num1, num2;

    cout << "Enter the first integer: ";
    cin >> num1;
    cout << "Enter the second integer: ";
    cin >> num2;
```

INPUT

finalshoa2.cpp > ...

```cpp
#include <iostream>
using namespace std;

int add(int a, int b) {
    return a + b;
}

int subtract(int a, int b) {
    return a - b;
}

int multiply(int a, int b) {
    return a * b;
}

double divide(int a, int b) {
    return (double) a / b;
}

int main() {
    int num1, num2;

    cout << "Enter the first integer: ";
    cin >> num1;
    cout << "Enter the second integer: ";
    cin >> num2;

    cout << endl;
    cout << "RESULTS OF OPERATIONS" << endl;
    cout << "Addition: " << add(num1, num2) << endl;
    cout << "Subtraction: " << subtract(num1, num2) << endl;
    cout << "Multiplication: " << multiply(num1, num2) << endl;

    if (num2 != 0) {
    cout << "Division: " << divide(num1, num2) << endl;
    }
    else {
        cout << "INVALID SECOND NUMBER" << endl;
    }


    return 0;
}

```

OUTPUT

Problems Output Debug Console Terminal Ports

```
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\finalshoa2.exe
Enter the first integer: 7
Enter the second integer: 4

RESULTS OF OPERATIONS
Addition: 11
Subtraction: 3
Multiplication: 28
Division: 1.75
PS C:\CPE11S4-DOMINGO>
```

EXPLANATION: This program is used to perform the four basic mathematical operations: addition, subtraction, multiplication, and division. Each operation has its own function, which makes the code more organized and easier to understand. The program asks the user to enter two integers, then displays the results of each operation using the given numbers. For division, it checks if the second number is not zero to avoid dividing by zero, and if it is zero, it displays an invalid number message instead. Overall, this program helps us understand how functions, user input, and conditional statements work together in C++.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

For each of the following, create the following c++ code:
1. Create a program in C++ that will convert a user-input Farenheight to Celcius and vice versa. Use functions in the program.

INPUT

finalshoa3.cpp > ...

```cpp
#include <iostream>
using namespace std;

double fahrenheit(float a) {
    return (float) (a*1.8) + 32;
}

double celcius(float a) {
    return (float) (a - 32) / 1.8;
}


int main() {
    float num1;
    char g;
    int d;

    cout << "Choose your temp: (1) Celcius -> Fahrenheit " << endl;
    cout << "                  (2) Fahrenheit -> Celcius " << endl;
    cout << "Decision: ";
    cin >> d;
    cout << endl;

    if (d == 1) {
        cout << "Celcius: ";
        cin >> num1;

        cout << "Celcius -> Fahrenheit: " << fahrenheit(num1) << endl;
    }
    else if (d == 2) {
        cout << "Fahrenheit: ";
        cin >> num1;

        cout << "Fahrenheit -> Celcius: " << celcius(num1) << endl;
    }
    else {cout << "INVALID NUMBER" << endl;}

    cout << "Continue? y/n: ";
    cin >> g;

    while (g != 'n') {
    cout << endl;
    cout << "Choose your temp: (1) Celcius -> Fahrenheit " << endl;
    cout << "                  (2) Fahrenheit -> Celcius " << endl;
    cout << "Decision: ";
    cin >> d;
    cout << endl;
    if (d == 1) {
        cout << "Celcius: ";
        cin >> num1;

        cout << "Celcius -> Fahrenheit: " << fahrenheit(num1) << endl;
    }
    else if (d == 2) {
        cout << "Fahrenheit: ";
        cin >> num1;

        cout << "Fahrenheit -> Celcius: " << celcius(num1) << endl;
    }
    else {cout << "INVALID NUMBER" << endl;}

    cout << "Continue? y/n: ";
    cin >> g;

    }
    cout << endl;
    cout << "Thank you.";

return 0;
}
```

OUTPUT



EXPLANATION: This program converts the temperature from Celsius to Fahrenheit or Fahrenheit to Celsius, depending on the user's choice. It uses two functions, fahrenheit() and celcius(), to calculate the converted temperature based on the formula. The program asks the user to choose a conversion, enter the temperature, and displays the result. I also added a while loop so the user can convert temperatures repeatedly without restarting the program. Once the user enters n, the loop stops and the program displays “Thank you.”

2. Create a program in C++ that will convert dollars into pesos and vice versa. Use functions in the program.

INPUT

finalshoa4.cpp > ...

```cpp
#include <iostream>
using namespace std;

double peso(float a) {
    return (float) a * 62.75;
}

double dollar(float a) {
    return (float) a * 0.016;
}


int main() {
    float num1;
    int d;

    cout << "Choose your conversion: (1) Dollars ($) -> Peso (PHP) " << endl;
    cout << "                        (2) Peso (PHP) -> Dollars ($) " << endl;
    cout << "Decision: ";
    cin >> d;
    cout << endl;

    if (d == 1) {
        cout << "Money in Dollars: ";
        cin >> num1;

        cout << "Dollars ($) -> Peso (PHP): PHP " << peso(num1) << endl;
    }
    else if (d == 2) {
        cout << "Money in Peso (PHP): ";
        cin >> num1;

        cout << "Peso (PHP) -> Dollars ($): $ " << dollar(num1) << endl;
    }
    else {cout << "INVALID NUMBER" << endl;}


return 0;
}
```

OUTPUT

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\finalshoa4.exe
Choose your conversion: (1) Dollars ($) -> Peso (PHP)
                        (2) Peso (PHP) -> Dollars ($)
Decision: 1

Money in Dollars: 100
Dollars ($) -> Peso (PHP): PHP 6275
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\finalshoa4.exe
Choose your conversion: (1) Dollars ($) -> Peso (PHP)
                        (2) Peso (PHP) -> Dollars ($)
Decision: 2

Money in Peso (PHP): 6275
Peso (PHP) -> Dollars ($): $ 100.4
PS C:\CPE11S4-DOMINGO>
```

EXPLANATION: This program converts dollars to Philippine pesos and pesos to dollars using two separate functions. The peso() function multiplies the dollar amount by 62.75 to convert it to pesos, while the dollar() function multiplies the peso amount by 0.016 to convert it to dollars. The program asks the user to choose which conversion they want and enter the amount of money. It then displays the converted amount based on the user's choice, and if the user enters an invalid option, it displays “INVALID NUMBER.” Overall, this program demonstrates how functions, user input, and conditional statements can be used to create a simple currency converter in C++.

3. Create a C++ program for a Shopping Cart Price Calculator. The program must allow the user to enter the prices and quantities of 5 different products. Store the prices and quantities in separate arrays and use for loops to process the data. Create separate functions to calculate the total cost of each product, compute the total bill, and determine the discount based on the following rules:
- 10% discount if the total bill is at least ₱3,000.
- 5% discount if the total bill is at least ₱1,500 but less than ₱3,000.
- No discount if the total bill is below ₱1,500.

Display each product's price, quantity, and subtotal, followed by the total bill, discount amount, and final amount to pay.

INPUT

finalshoa5.cpp > ...

```cpp
#include <iostream>
using namespace std;

struct Items {
    float price;
    int quantity;
};

float product(float a, float b) {
    return (float) a*b;
}
float disc(float a, float b) {
    return (a) - (a*b);
}

int main() {
    Items cart;
    int i = 1, t = 1, j = 1, k = 1;
    int quant[5];
    float price[5];
    float prodprice[5];

    for (int g = 0; g < 5; g++) {
        cout << "Item " << i << " Price: PHP ";
        cin >> cart.price;
        cout << "Item " << i << " Quantity: ";
        cin >> cart.quantity;

        price[g] = cart.price;
        quant[g] = cart.quantity;
        i++;

    }
    float totalbill = 0;
    for (int w = 0; w < 5; w++) {
    prodprice[w] = product(price[w], quant[w]);

    }
    for (int y = 0; y < 5; y++) {
        totalbill += prodprice[y];
    }

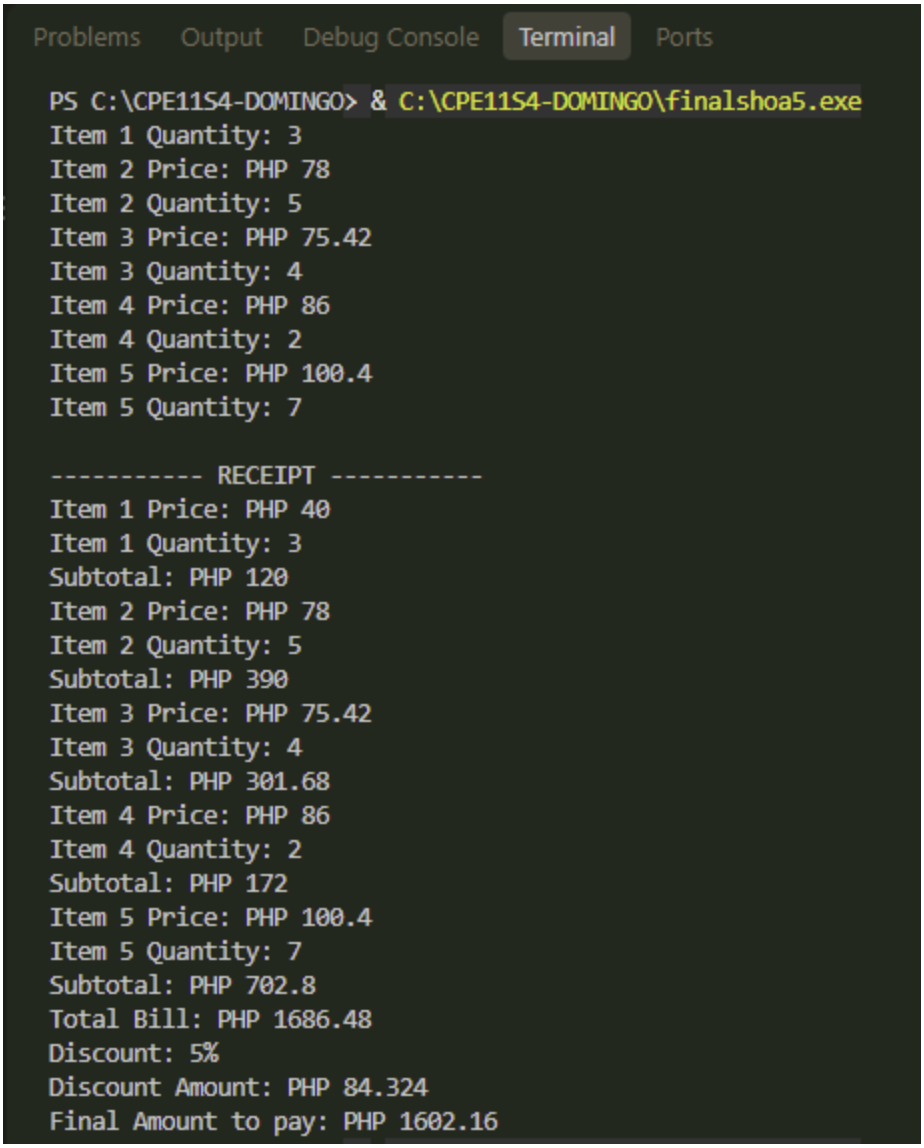
```

```
    if (totalbill >= 3000) {
        cout << endl;
        cout << "---------- RECEIPT ---------- " << endl;
        for (int e = 0; e < 5; e++) {
            cout << "Item " << t << " Price: PHP " << price[e] << endl;
            cout << "Item " << t << " Quantity: " << quant[e] << endl;
            cout << "Subtotal: PHP " << prodprice[e] << endl;
            t++;
        }
            cout << "Total Bill: PHP " << totalbill << endl;
            cout << "Discount: 10%" << endl;
            cout << "Discount Amount: PHP " << product(totalbill, 0.10) << endl;
            cout << "Final Amount to pay: PHP " << disc(totalbill, 0.10) << endl;

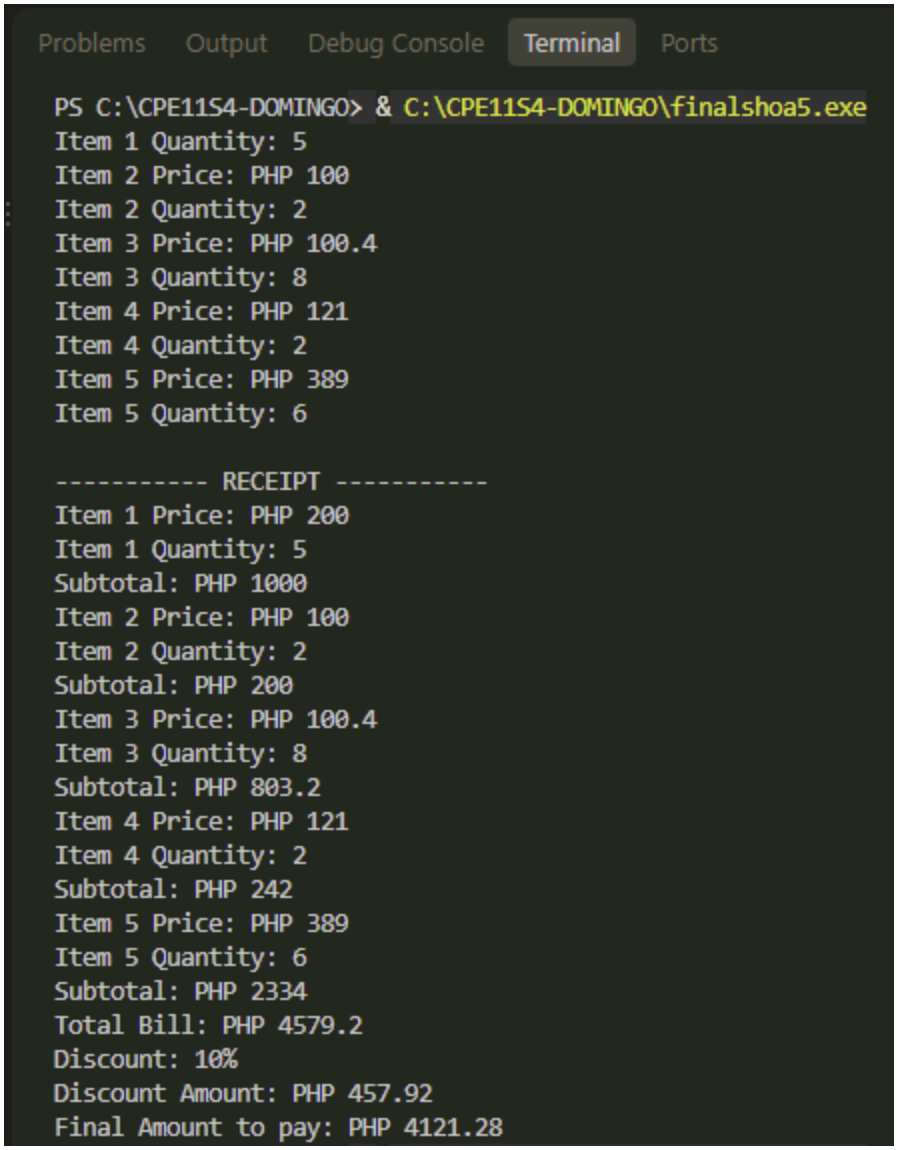
    }
    else if (totalbill >= 1500 && totalbill < 3000) {
        cout << endl;
        cout << "---------- RECEIPT ---------- " << endl;
        for (int r = 0; r < 5; r++) {
            cout << "Item " << j << " Price: PHP " << price[r] << endl;
            cout << "Item " << j << " Quantity: " << quant[r] << endl;
            cout << "Subtotal: PHP " << prodprice[r] << endl;
            j++;
        }
            cout << "Total Bill: PHP " << totalbill << endl;
            cout << "Discount: 5%" << endl;
            cout << "Discount Amount: PHP " << product(totalbill, 0.05) << endl;
            cout << "Final Amount to pay: PHP " << disc(totalbill, 0.05) << endl;
    }
    else {
        cout << endl;
        cout << "---------- RECEIPT ---------- " << endl;
        for (int h = 0; h < 5; h++) {
            cout << "Item " << k << " Price: PHP " << price[h] << endl;
            cout << "Item " << k << " Quantity: " << quant[h] << endl;
            cout << "Subtotal: PHP " << prodprice[h] << endl;
            k++;
        }
            cout << "Total Bill: PHP " << totalbill << endl;
            cout << "Discount: NO DISCOUNT." << endl;
            cout << "Final Amount to pay: PHP " << totalbill << endl;
    }


return 0;
}
```

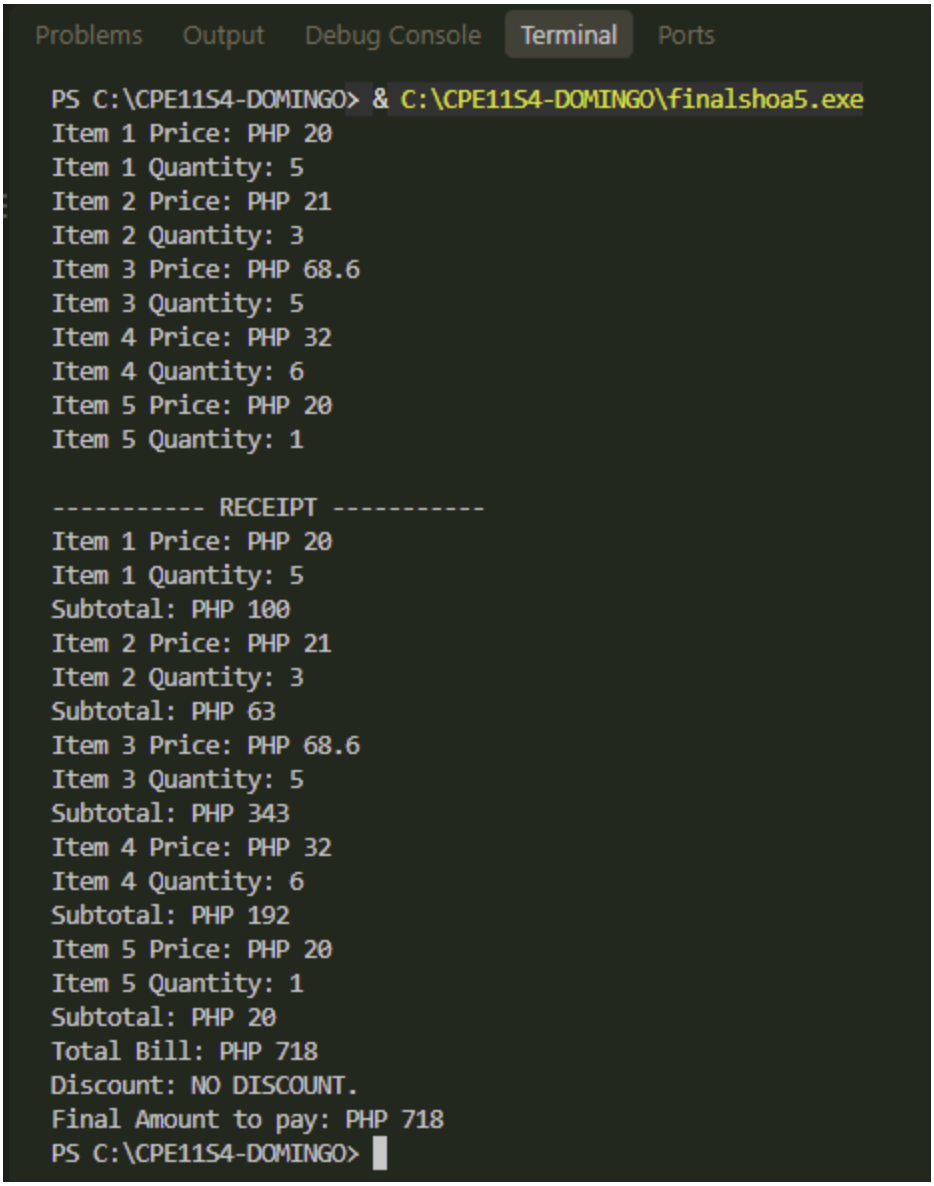
OUTPUT


Problems
Output
Debug Console
Terminal
Ports
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\finalshoa5.exe
Item 1 Quantity: 3
Item 2 Price: PHP 78
Item 2 Quantity: 5
Item 3 Price: PHP 75.42
Item 3 Quantity: 4
Item 4 Price: PHP 86
Item 4 Quantity: 2
Item 5 Price: PHP 100.4
Item 5 Quantity: 7
---------- RECEIPT ----------
Item 1 Price: PHP 40
Item 1 Quantity: 3
Subtotal: PHP 120
Item 2 Price: PHP 78
Item 2 Quantity: 5
Subtotal: PHP 390
Item 3 Price: PHP 75.42
Item 3 Quantity: 4
Subtotal: PHP 301.68
Item 4 Price: PHP 86
Item 4 Quantity: 2
Subtotal: PHP 172
Item 5 Price: PHP 100.4
Item 5 Quantity: 7
Subtotal: PHP 702.8
Total Bill: PHP 1686.48
Discount: 5%
Discount Amount: PHP 84.324
Final Amount to pay: PHP 1602.16

OUTPUT


Problems
Output
Debug Console
Terminal
Ports
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\finalshoa5.exe
Item 1 Quantity: 5
Item 2 Price: PHP 100
Item 2 Quantity: 2
Item 3 Price: PHP 100.4
Item 3 Quantity: 8
Item 4 Price: PHP 121
Item 4 Quantity: 2
Item 5 Price: PHP 389
Item 5 Quantity: 6
---------- RECEIPT ----------
Item 1 Price: PHP 200
Item 1 Quantity: 5
Subtotal: PHP 1000
Item 2 Price: PHP 100
Item 2 Quantity: 2
Subtotal: PHP 200
Item 3 Price: PHP 100.4
Item 3 Quantity: 8
Subtotal: PHP 803.2
Item 4 Price: PHP 121
Item 4 Quantity: 2
Subtotal: PHP 242
Item 5 Price: PHP 389
Item 5 Quantity: 6
Subtotal: PHP 2334
Total Bill: PHP 4579.2
Discount: 10%
Discount Amount: PHP 457.92
Final Amount to pay: PHP 4121.28

OUTPUT



EXPLANATION: This program acts as a simple shopping cart that calculates the subtotal for five items based on their prices and quantities. It uses a struct to store each item's price and quantity, while arrays keep track of the information for all five items. After getting the total bill, the program checks which discount applies: 10% for bills worth PHP 3,000 or more, 5% for bills from PHP 1,500 to below PHP 3,000, and no discount for lower amounts. The product() function is used to calculate the subtotal and discount amount, while the disc() function calculates the final amount after deducting the discount. Lastly, it displays a receipt containing each item's details, the total bill, the discount, and the final amount to pay.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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